A1.1 Domains of integer powers

For $b \neq 0, n \in \mathbb{Z}_{\geq 0}$, define

$$b^0 = 1, \quad b^n = b \cdot b^{n-1}, \quad b^{-n} = 1/b^n.$$

At $b = 0$, positive integer powers exist, while 0^0 and negative powers require contextual conventions or exclusion. The laws

$$b^{m+n} = b^m b^n, \quad (b^m)^n = b^{mn}, \quad (bc)^n = b^n c^n$$

require all appearing powers to be defined. There is no universal prior evaluation of 0^0; conventions for limits, series, and combinatorial counting are stated where needed.

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A1.2 Rational powers and principal roots

For $b > 0, p \in \mathbb{Z}, q \in \mathbb{Z}_{>0}$, define $b^{p/q} = (b^{1/q})^p$, where $b^{1/q}$ is the unique positive r satisfying $r^q = b$. Fraction representation does not affect the result, for example $b^{2/4} = b^{1/2}$.

For negative bases, reduce the fraction first. A real rational power is defined through an odd root only when the reduced denominator is odd. Thus $(-8)^{2/6} = (-8)^{1/3} = -2$ is valid despite the unreduced even denominator, while $(-4)^{1/2}$ is not real. The expressions $(x^2)^{1/4}$ and $x^{1/2}$ have different domains; nested-power laws cannot be applied unconditionally to negative bases.

The principal-root convention is part of well-definedness. For $b > 0$, $r^2 = b$ has two real solutions, but its principal square root is the positive one. At zero it is zero.

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A1.3 Defining exp by a power series

For every real x, define

$$\exp(x) = E(x) = \sum_{n=0}^{\infty} x^n/n!.$$

The ratio test gives infinite radius of convergence and absolute convergence on bounded intervals. Termwise differentiation gives

$$E'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{m=0}^{\infty} \frac{x^m}{m!} = E(x),$$

with $E(0) = 1$. Thus E solves $y' = y, y(0) = 1$. After establishing the exponential law, any other solution has $(y(x)E(-x))' = 0$ and initial value one, so $y = E$, proving uniqueness. Infinite radius does not make E a polynomial merely because each term is polynomial.

A1.4 Proving the exponential law

Absolute convergence permits multiplication and rearrangement:

$$\begin{aligned} E(x)E(y) &= \sum_{m,k\geq 0} \frac{x^m y^k}{m!k!} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^{n} \binom{n}{m} x^m y^{n-m} \\ &= \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!} = E(x+y). \end{aligned}$$

Taking $y = -x$ gives $E(x)E(-x) = 1$, hence nonvanishing. Since $E(x) = E(x/2)^2$, it is positive and $E(-x) = 1/E(x)$.

Define $e = E(1)$. Induction and roots give $E(r) = e^r$ for rational r; continuity extends this to reals. The binomial expansion proves the identity, and absolute convergence on compact intervals justifies rearrangement.

A1.5 Logarithm as an inverse

Since $E' = E > 0$, E is strictly increasing and injective. For nonnegative x, $E(x) \geq 1 + x$, so it tends to infinity. The reciprocal identity makes $E(x) \to 0$ as $x \to -\infty$. Continuity and the intermediate value theorem therefore give image $(0, \infty)$, and unique inverse functions

$$\exp : \mathbb{R} \to (0, \infty), \qquad \ln : (0, \infty) \to \mathbb{R},$$
$$\ln(e^x) = x, \qquad e^{\ln y} = y.$$

The real logarithm requires positive arguments. Complex logarithms require branch choices beyond this single-valued real definition.

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A1.6 Logarithm laws and change of base

For $b > 0$, define every real power by $b^r = E(r \ln b)$. Uniqueness of positive roots makes this agree with the earlier rational definition. The exponential law and inverse give, for $a, b > 0, r \in \mathbb{R}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln(a/b) = \ln a - \ln b, \quad \ln(a^r) = r \ln a.$$

They also prove positive-base real-exponent laws:

$$b^{x+y} = E((x + y) \ln b) = b^x b^y,$$
$$(ab)^x = E(x(\ln a + \ln b)) = a^x b^x,$$
$$(b^x)^y = E(y \ln(E(x \ln b))) = E(xy \ln b) = b^{xy}.$$

The quotient law follows from $b^{-x} = 1/b^x$. All require positive bases and the established inverse.

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Change of base and asymptotic comparison

For $b > 0, b \neq 1$, define $\log_b x = \ln x / \ln b$. Fixed $a, b > 1$ give

$$\log_a x = \frac{\log_b x}{\log_b a},$$

a positive constant factor, hence Θ-equivalence. For a base between zero and one, logarithms are negative at $x > 1$, so positive-cost comparison cannot use the same positive-factor argument directly. Compare absolute values, $|\log_a x| = \Theta(|\log_b x|)$, or retain signs explicitly.

Equal asymptotic classes do not imply equal implementation cost, error, or overflow behavior. The product law does not extend to sums: for $a = b = 1$, $\ln(a + b) = \ln 2$ but $\ln a + \ln b = 0$. The two expressions can nevertheless coincide for particular inputs.

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Logarithm notation: where does the exponent go?

Fix a base $b > 1$, $n > 1$, and a real constant k. These expressions mean different things:

$$\log_b^k n = (\log_b n)^k, \qquad \log_b(n^k) = k \log_b n.$$

For $b = 2, n = 16, k = 2$, the first is $4^2 = 16$, while the second is $\log_2 256 = 8$. Write parentheses when the convention might be unclear.

The iterated logarithm $\log_b^* n$ is not a power: it counts the logarithms needed until the result is at most 1. With this convention, $\log_2^* 16 = 3$, since $16 \to 4 \to 2 \to 1$.

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Exchanging a base and an argument

For $a, c > 0$ and $b > 0, b \neq 1$, the definition gives

$$a^{\log_b c} = \exp\left(\frac{\ln a \, \ln c}{\ln b}\right) = c^{\log_b a}.$$

For $n > 0$, using base 2 explicitly,

$$4^{\log_2 n} = n^{\log_2 4} = n^2, \qquad (\sqrt{2})^{\log_2 n} = n^{1/2} = \sqrt{n}.$$

Changing bases does not change a value when the conversion factor is included. In asymptotic comparisons, logarithms of fixed bases greater than 1 differ by a positive constant. Check the local convention whenever the base of $\log$ is omitted.

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A1.7 Derivatives, growth, and boundaries

The inverse derivative gives $(\ln x)' = 1/x$, and $(b^x)' = (\ln b) b^x$. For every real x, $e^x \geq 1 + x$. Nonnegative x permits stronger polynomial bounds from the series. In particular, for fixed n, $x^n / e^x \to 0$, so $x^n = o(e^x)$.

For $b = 0.5$, b^x decreases; it increases only for $b > 1$. State the base range. Mathematical identities need not behave equally in floating point. With $a = b = 10^{200}$, double-precision ab overflows while $\log(a) + \log(b)$ can remain finite. Numerical strategy depends on the input range.

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A1.8 Existence and representation independence of rational powers

For $b > 0$ and positive integer q, the function t^q is continuous and strictly increasing on nonnegative reals. For $t > s \geq 0$,

$$t^q - s^q = (t - s)(t^{q-1} + \cdots + s^{q-1}) > 0.$$

The intermediate value theorem between zero and sufficiently large t gives a unique positive q-th root of b.

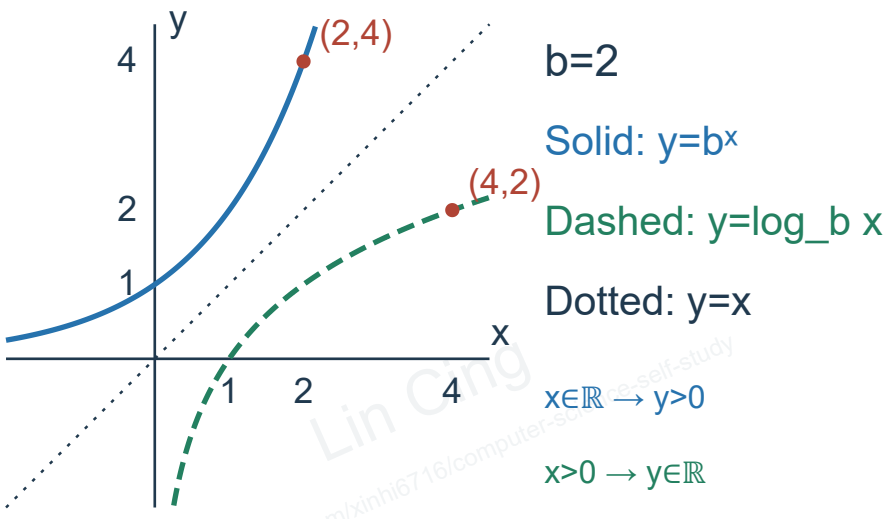
If $p/q = p'/q'$, raising the two positive candidates to power qq' gives $b^{pq'} = b^{p'q}$. Injectivity of a positive integer power forces equality. Reciprocals handle negative numerators. Thus the definition is independent of fractional representation.

For negative bases, even defined nested powers may fail the usual law:

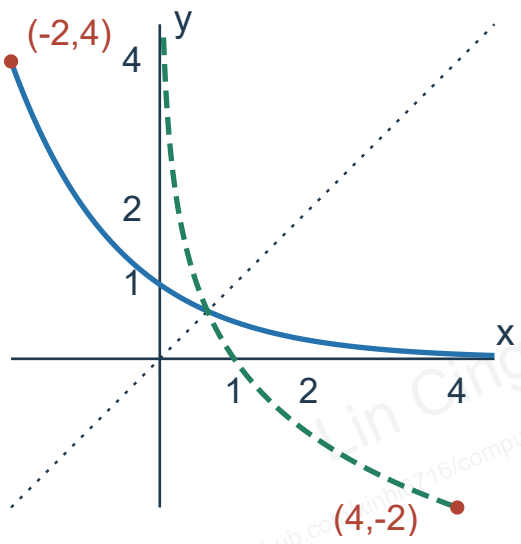
$$((-8)^2)^{1/2} = 8, \qquad (-8)^{2(1/2)} = -8.$$

Existence of both expressions alone does not extend the positive-base law.

A1.9 Inverse graphs and base-dependent monotonicity



Inverse graphs for a base below 1



b=0.5

Solid: $y=b^x$

Dashed: $y=\log_b x$

Dotted: $y=x$

$x\in\mathbb{R} \to y>0$

$x>0 \to y\in\mathbb{R}$

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Monotonicity and inequalities

Inversion exchanges (x, b^x) and (b^x, x), reflecting the graph in $y = x$. Both functions increase for $b > 1$, where $\ln b > 0$, and decrease for $0 < b < 1$, where $\ln b < 0$. Both remain one-to-one. The exponential range and logarithm domain are the positive reals.

Because the base is below one,

$$\log_{1/2} x \geq 2 \iff 0 < x \leq (1/2)^2 = 1/4.$$

The inverse reverses the inequality. Substituting $x = 1/8$ gives $3 \geq 2$, while $x = 1$ gives zero and fails.

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A1.10 Quantified growth proofs

For fixed integer $k \geq 0, x > 0$, one series term gives

$$e^x \geq \frac{x^{k+1}}{(k+1)!} \Rightarrow 0 \leq \frac{x^k}{e^x} \leq \frac{(k+1)!}{x} \to 0.$$

For fixed real $r \geq 0$, choose integer $k \geq r$; when $x \geq 1$, $x^r \leq x^k$, proving $x^r = o(e^x)$. For fixed $b > 1$, substituting $t = x \ln b$ gives $x^r = o(b^x)$.

For fixed $\varepsilon > 0$, put $u = \varepsilon \ln x \to \infty$. Then

$$\frac{\ln x}{x^\varepsilon} = \frac{1}{\varepsilon} \frac{u}{e^u} \to 0.$$

Thus $\ln x = o(x^\varepsilon)$. Fixing ε, r, b matters; the same conclusion need not hold if b approaches one with x.

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A1.11 Halving rounds, bit lengths, and rounding

A positive integer's binary length is $\lfloor \log_2 n \rfloor + 1$. The unique k satisfying $2^k \leq n < 2^{k+1}$ indexes its highest significant bit.

n	Bounding powers	Binary	Length
8	$8 \leq 8 < 16$	1000	4
13	$8 \leq 13 < 16$	1101	4
16	$16 \leq 16 < 32$	10000	5

Replacing workload n by $\lceil n/2 \rceil$ each round until one needs exactly $\lceil \log_2 n \rceil$ rounds for $n \geq 1$, zero at one. After t rounds the value is $\lceil n/2^t \rceil$, by induction using $\lceil \lceil z \rceil / 2 \rceil = \lceil z/2 \rceil$. It first reaches one iff $2^t \geq n$. This differs from bit length, especially at powers of two.

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A1.12 Logarithms within numerical range

For $L = \ln(e^a + e^b)$, set $m = \max(a, b)$:

$$L = m + \ln(e^{a-m} + e^{b-m}).$$

Both exponents are nonpositive and one exponential equals one. The sum lies in $[1, 2]$, avoiding overflow in e^a. For $a = 1000, b = 999$, $L = 1000 + \ln(1 + e^{-1}) \approx 1000.313262$. This applies a real identity with input-range control, not the false splitting of a logarithm of a sum.

For the exponential series on $|x| \leq M$, $|x^n/n!| \leq M^n/n!$, with a convergent majorant. The Weierstrass M-test gives uniform convergence. The derivative series has the same bounded-interval estimate. Termwise differentiation uses a power-series theorem, not merely differentiability of individual terms.

Separate exercise and solution PDFs provide domain checks, counterexamples, and quantitative bounds.

Key ideas

Check the base and argument before applying exponent or logarithm laws. Examine the original domain when handling nested powers, decreasing logarithms, and floating-point overflow.

References and attribution

Lin Cing, handwritten data structures notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Appendix 1.

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